

Project Design Phase
Proposed Solution Template

Date	11 July 2026
SID	SWUID2026230011
Project Name	Customer Care Registry (MERN Stack)
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Customer support requests currently come in through scattered channels — phone calls, emails, WhatsApp messages — with no central log. Customers must repeat their issue to multiple people, agents lose track of open complaints, and managers have no reliable way to measure resolution time, recurring issues, or agent performance.
2.	Idea / Solution Description	A Customer Care Registry web application (MERN stack) that gives customers a single place to register, raise a ticket with category/description/attachment, and track its status in real time. Support agents get a dashboard of assigned tickets to update and resolve, while administrators get an analytics dashboard for ticket volume, resolution time, and agent performance, with automatic SLA-based escalation for unresolved tickets.
3.	Novelty / Uniqueness	Combines a lightweight, easy-to-adopt ticketing flow (unlike heavy enterprise helpdesk suites) with built-in SLA escalation, real-time status tracking via Socket.io, and an integrated feedback/rating loop that feeds directly into agent-performance analytics — all in one role-based system for customers, agents, and admins.
4.	Social Impact / Customer Satisfaction	Reduces customer frustration by giving them visibility into their complaint's status and a single point of contact instead of repeated follow-ups. Faster, more accountable resolution builds trust, and the feedback loop lets businesses continuously improve service quality and training based on real customer input.
5.	Business Model (Revenue Model)	SaaS subscription for small & medium businesses, tiered by number of agents and monthly ticket volume (Basic / Pro / Enterprise plans). Optional add-ons include advanced analytics, WhatsApp/email channel integration, and white-labelling for larger clients.
6.	Scalability of the Solution	Built on the MERN stack with a stateless REST API and MongoDB Atlas, the system scales horizontally by adding backend instances behind a load balancer. Modular controllers (auth, ticket, feedback, report) allow new channels (chatbot, WhatsApp API) or features (AI-based auto-categorization) to be added without disrupting the existing architecture.